

PROJECT TITLE

AI-Powered Intelligent Endpoint Monitoring and Automated IT Incident Management System

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Domain,

Artificial Intelligence and Machine Learning (AI/ML).

Title,

INTELLIGENT ENDPOINT AI: CONTINUOUS SYSTEM MONITORING AND AUTOMATED INCIDENT MANAGEMENT FOR PREDICTIVE IT OPERATIONS

PICO

Problem : Enterprise IT endpoints and incidents

Intervention : AI-assisted telemetry analysis + XGBoost

Comparison : Random Forest / LightGBM or conventional monitoring

Output : Higher accuracy and lower false-positive rate

Introduction

- This project introduces an AI-Powered Intelligent Endpoint Monitoring and Automated IT Incident Management System to improve the monitoring and management of organizational computer systems.
- The system continuously monitors CPU, RAM, disk health, temperature, network activity, application crashes, and system logs across multiple endpoints.
- Artificial Intelligence and Machine Learning techniques are used to detect abnormal behavior, identify possible root causes, and predict potential system failures.
- The system automatically generates incident tickets, detects duplicate issues, and recommends troubleshooting solutions using historical incidents and a knowledge base.
- The proposed framework enables proactive IT support, reducing system downtime, manual effort, and incident resolution time while improving overall IT operational efficiency.

Related Works,

1. Traditional IT infrastructure monitoring systems such as Nagios and Zabbix continuously track CPU, memory, disk, network, and system performance using predefined thresholds, but they have limited intelligent diagnosis capabilities.
2. AI-based anomaly detection approaches have been developed to identify unusual system behavior from monitoring data using Machine Learning algorithms such as Isolation Forest, One-Class SVM, and clustering techniques.
3. Automated IT Service Management (ITSM) systems focus on incident detection, ticket generation, classification, prioritization, and resolution, but many existing solutions still require significant manual intervention.
4. Recent research on AIOps (Artificial Intelligence for IT Operations) combines monitoring data, log analysis, Machine Learning, and automation to detect incidents, identify root causes, and predict potential failures.
5. The proposed system extends these approaches by integrating real-time endpoint monitoring, AI-based anomaly detection, automated incident management, duplicate incident detection, root-cause analysis, and predictive maintenance into a unified platform.

Derivative From Based Paper,

- The proposed project is derived from the AIOps and AI-based IT infrastructure monitoring approach, where Artificial Intelligence is used to analyze large volumes of system and operational data.
- Existing research on AI-based anomaly detection provides the foundation for identifying abnormal behavior in endpoint performance and system logs.
- The concept of automated incident management is adapted to automatically create, classify, prioritize, and track IT support incidents.
- The project further extends these approaches by integrating root-cause analysis, duplicate incident detection, solution recommendation, and predictive maintenance.
- Thus, the proposed system combines existing AI-based monitoring concepts into a unified intelligent endpoint monitoring and automated IT incident management platform.

Reference

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- Jia, P., Cai, S., Ooi, B.C., Wang, P., Xiong, Y., 2023. Robust and transferable log-based anomaly detection. Proceedings of the ACM on Management of Data 1(1), 64. <https://doi.org/10.1145/3588918>.

THANK YOU